THE EYE

LEARNING OBJECTIVES

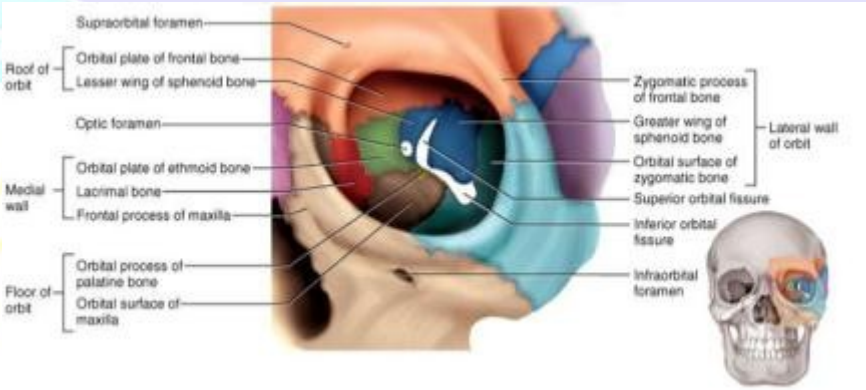
At the end of class the student should be able to:

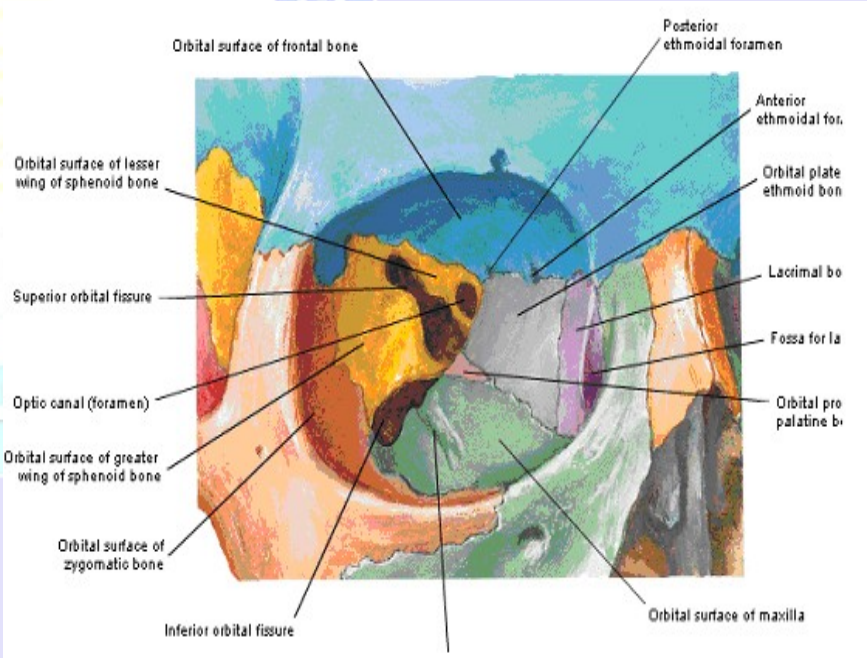
- Discuss the bony orbit and its boundaries
- Discuss the extraocular muscles and their nerve supply
- Discuss the actions of extraocular muscles
- Describe the eye ball and its coats.

THE ORBIT

- Orbit is a bony cavity shaped like a four-sided pyramid
- Lying on each side of the root of the nose
- Lodges the eyeball.

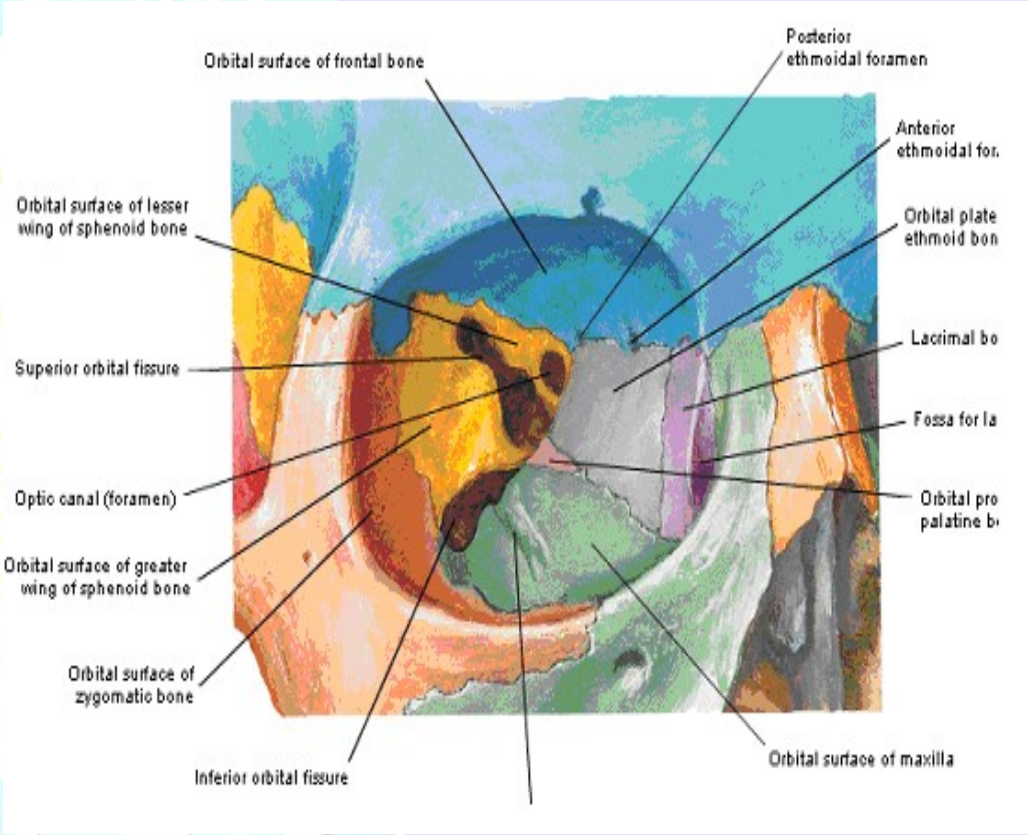
- **Boundaries of orbit**
- **An apex** is at the posterior end
- **A base** is orbital margins
- **4Walls** Roof, Floor, Lateral & Medial walls





ROOF

- **Formed by**
- Orbital part of the frontal bone
- Lesser wing of the sphenoid bone
- The roof separates the orbit from the anterior cranial fossa.

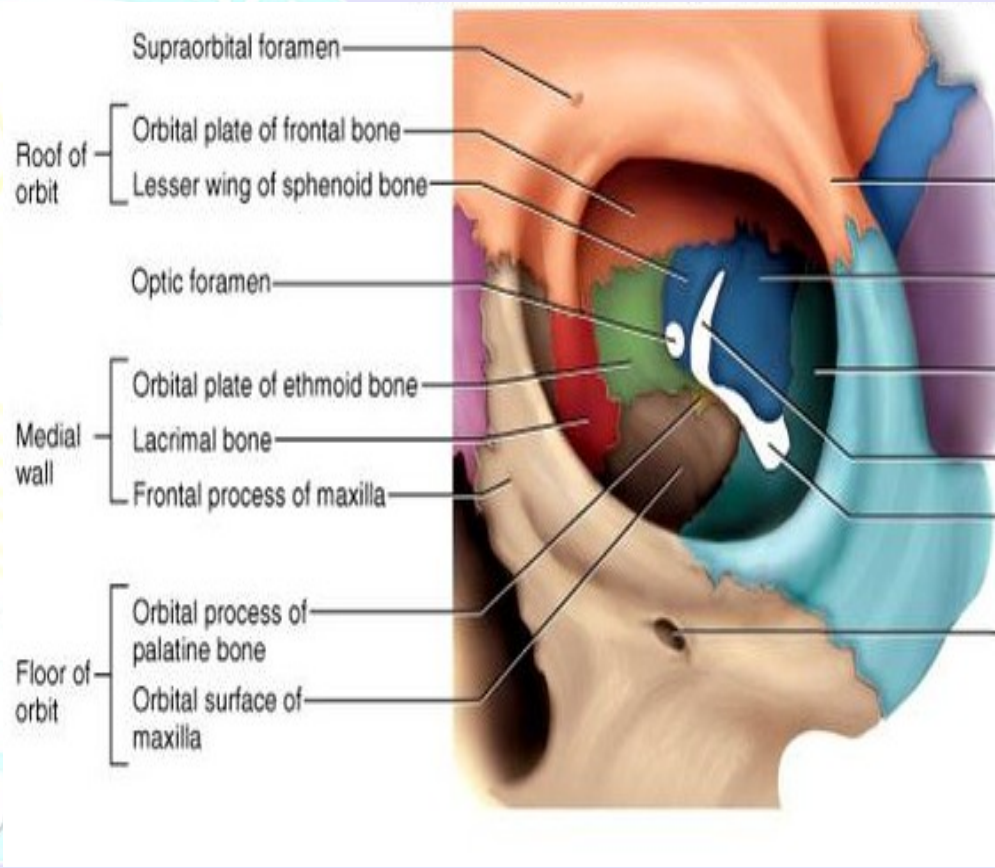


- **FEATURES:**
- Lacrimal Fossa
- Optic Canal
- Trochlear fossa

MEDIAL WALL

- 5 cm long. Separates orbit from the ethmoidal ear cells.
- Formed by (before backwards).
- Frontal process of the maxilla.
- Lacrimal bone
- Orbital plate of the ethmoid bone.
- Body of the sphenoid bone.

- **FEATURES:**
- Fossa for the Lacrimal sac….b/w ant. And post. Lacrimal crest
- Anterior Ethmoidal foramina…….24mm behind the ant. lacrimal crest
- Posterior ethmoidal foramina…….12mm behind this
- Optic nerve emerges 6mm further back



LATERAL WALL

- Thickest wall, 5cm long

Formed :

- Anteriorly by orbital suface of the frontal process of the zygomatic bone

(Separates orbit from temporal fossa)

- Posteriorly by ant. Surface of the greater wing of the sphenoid bone

(Separates orbit from middle cranial fossa

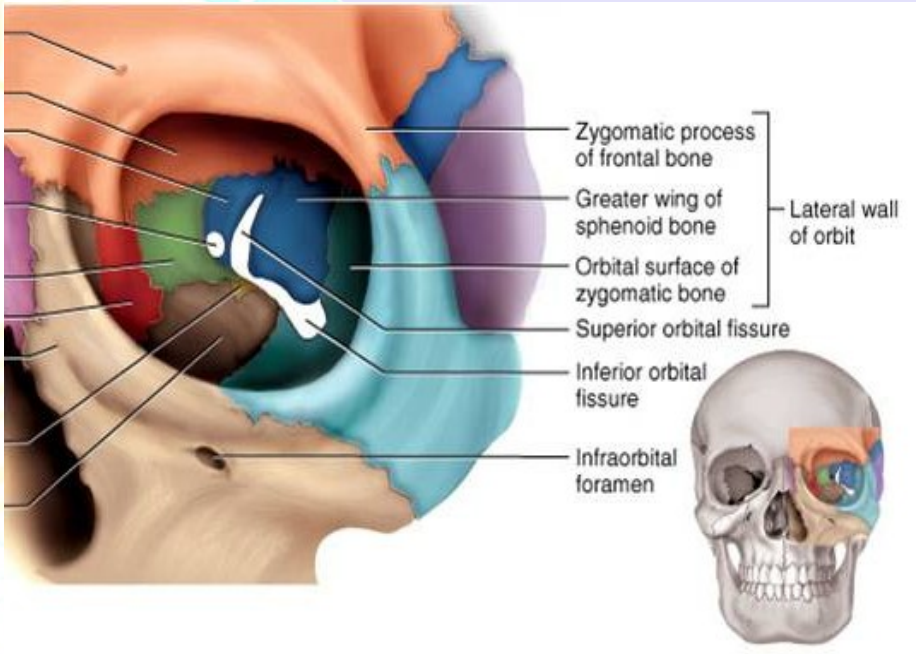
FEATURES:

- **Superior orbital fissure**…gap b/w lateral wall and roof

(leads to middle cranial fossa)

- **Inferior orbital fissure**…..gap b/w lateral wall & floor

(Leads to pterygopalatine & infratemporal fossae)



FLOOR

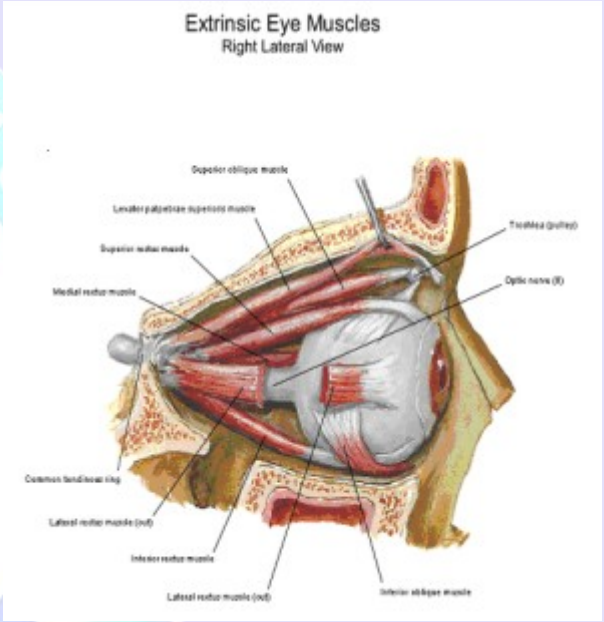
Formed by:

- Orbital surface of the maxilla
- Lower part of the orbital surfaces of the zygomatic bone
- Orbital process of the palatine bone
- Separates the orbit from the maxillary sinus

- **ORBITAL MARGINS** Supraorbital margin by frontal bone
- Infraorbital margin by zygomatic bone & maxilla
- Medial margin by ant. Lacrimal crest and frontal bone
- Lateral margin by frontal bone and zygomatic

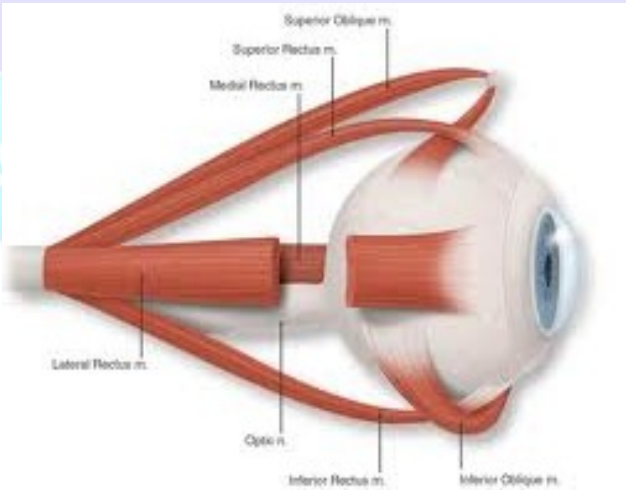
EXTRAOCULAR MUSCLES

- Eyeball is moved by extrinsic or the extraocular muscles: Four recti, Two obliqui, Levator palpebrae superioris
- **THE RECTI MUSCLES** The recti muscles arise from a common tendinous ring
- ✓ Superior & medial recti also arise from the dural sheath of the optic nerve
- ✓ Lateral rectus also arises from the orbital surface of the greater wing of the sphenoid bone
- ✓ All pierce the fascial sheath of the eyeball to get inserted into the sclera



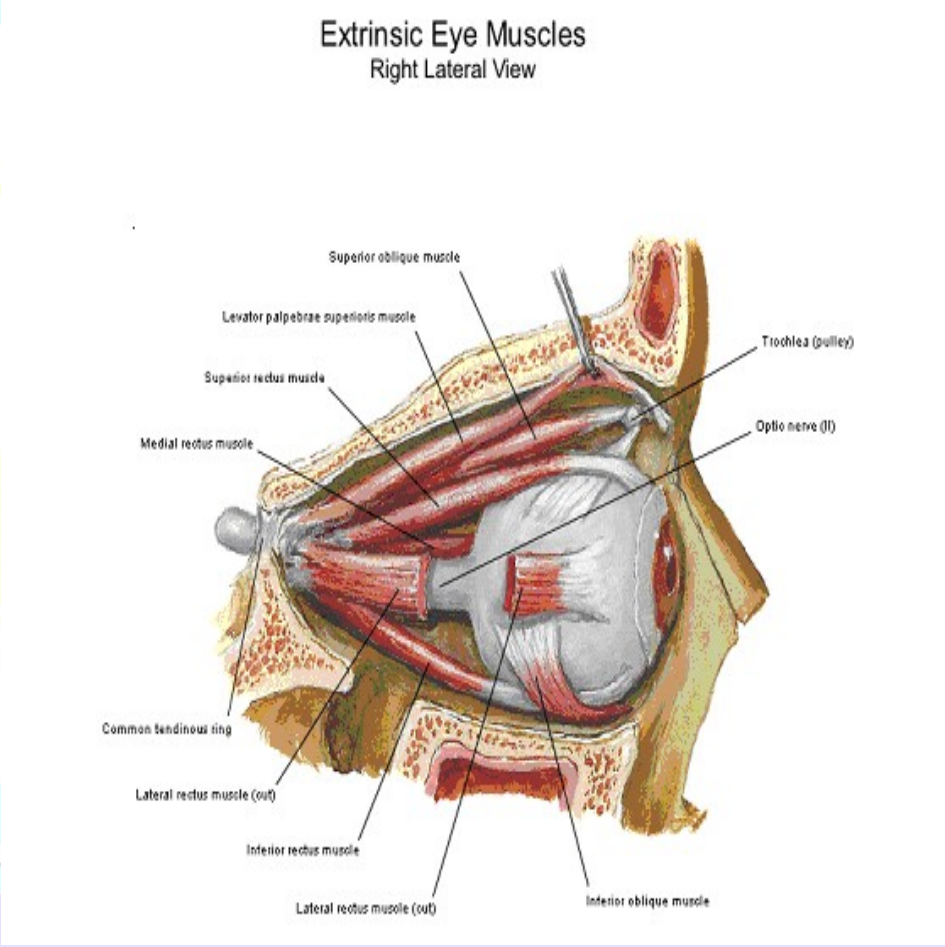
OBLIQUE MUSCLES

- **Superior oblique** arises from body of the sphenoid and
- gets inserted into the posterolateral quadrant of the sclera
- **Inferior Oblique** orbital surface of the maxilla (origin)
- Posterioinferior quadrant of the sclera (insertion)



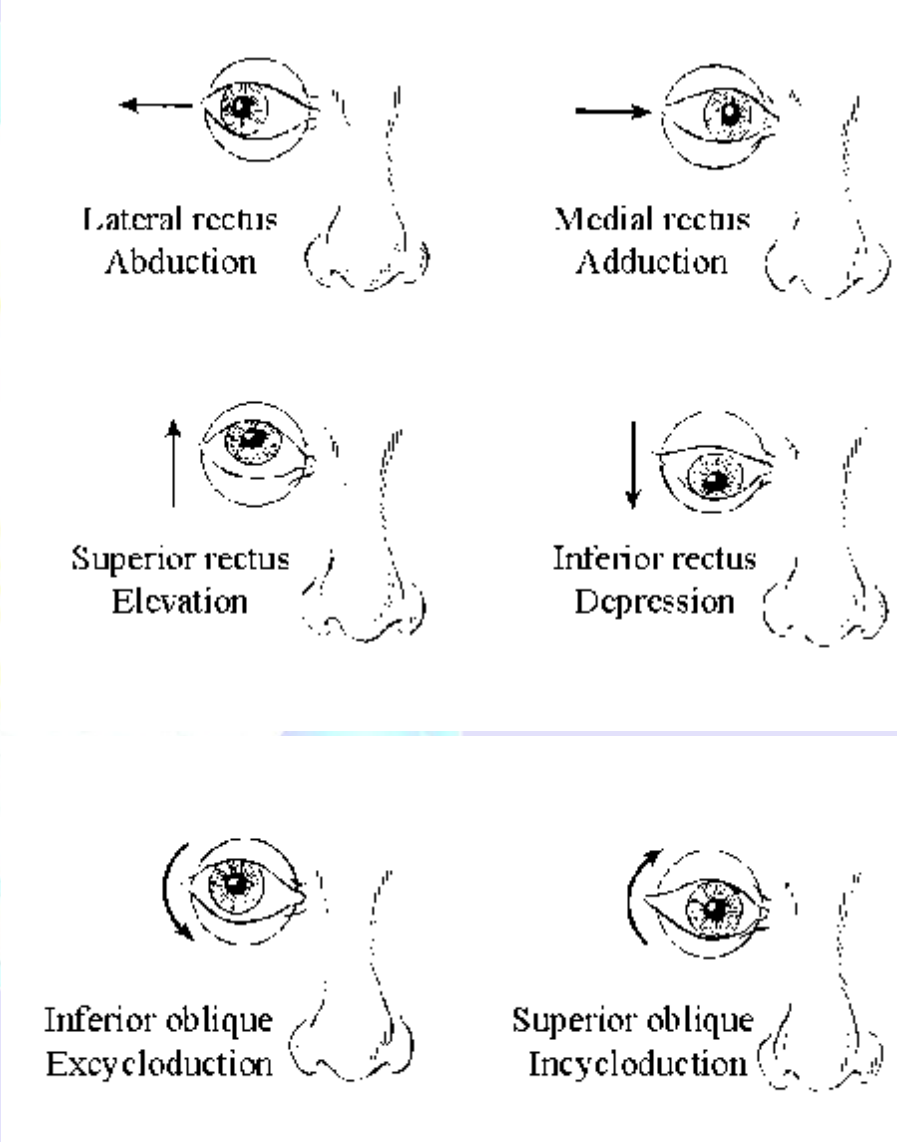
LEVATOR PALPABRAE SUPERIORIS

- ✓ Arises from undersurface of the lesser wing of the sphenoid
- ✓ Splits into two lamellae
- ✓ Superior (voluntary) inserts into ant surface of superior tarsus and skin of upper eyelid
- ✓ Inferior (involuntary) inserts into upper margin of superior tarsus

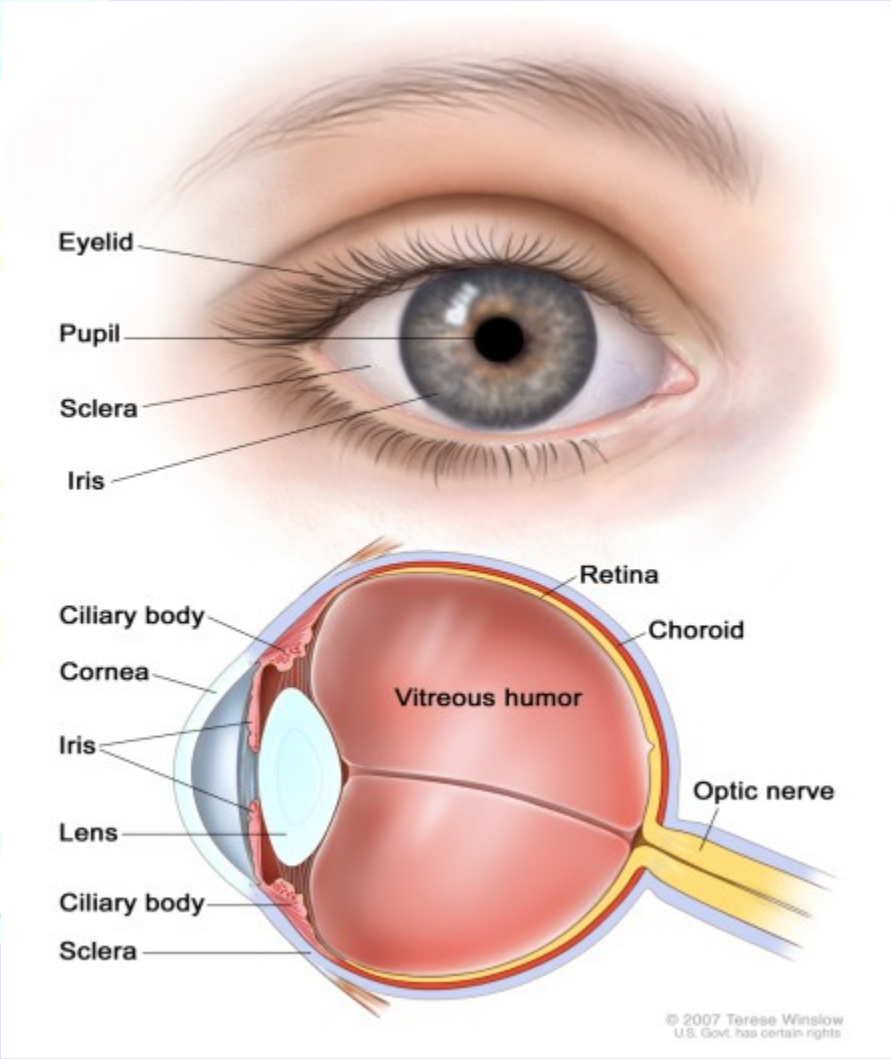


- **NERVE SUPPLY**
- (LR6 SO4)
- Lateral Rectus: abducent nerve
- Superior oblique: Trochlear nerve
- Remaining muscles: occulomotor nerve

- ACTIONS
- **Levator Palpabrae Superioris**: Elevation of upper eyelid
- **Superior rectus**: Upward rotation, medial rotation, intortion
- **Inferior rectus**: Downward rotation medial rotation extortion
- **Medial rectus:** Medial rotation
- **Lateral rectus :** Lateral rotation
- **Superior oblique:** Downward rotation, Lateral rotation
- **Inferior oblique:** Upward rotation Lateral rotation, Extortion



- The is a complex and highly developed **photosensitive organ**, that permits an accurate analysis of the light intensity, form and colour which are reflected from objects.

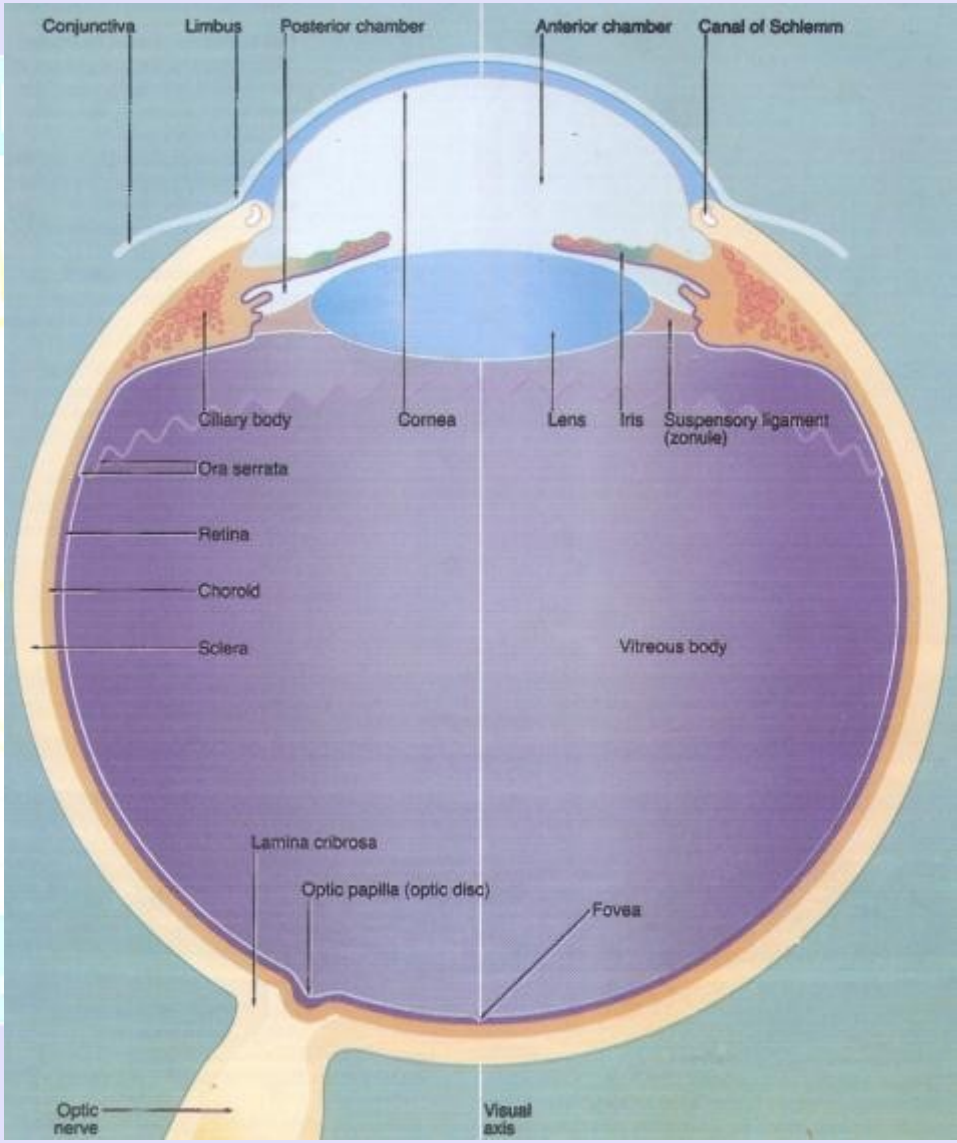


- Wall of the eyeball is composed of **three concentric coats**.

1.External (Fibrous) coat: that comprises of, sclera and cornea.

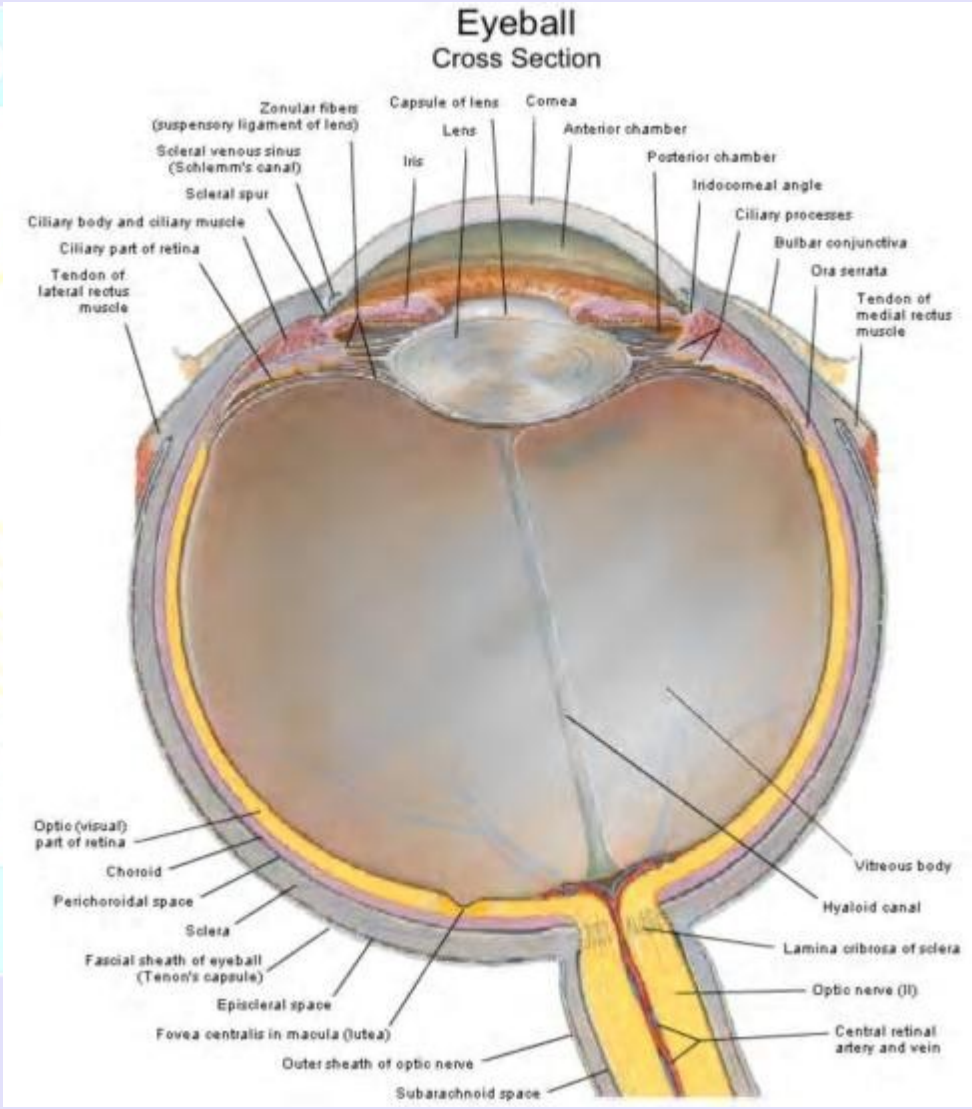
2.Middle (Vascular) coat: also called uvea or uveal tract that consists of choroid, ciliary body and iris.

3.Internal (nervous) coat: i.e. retina.



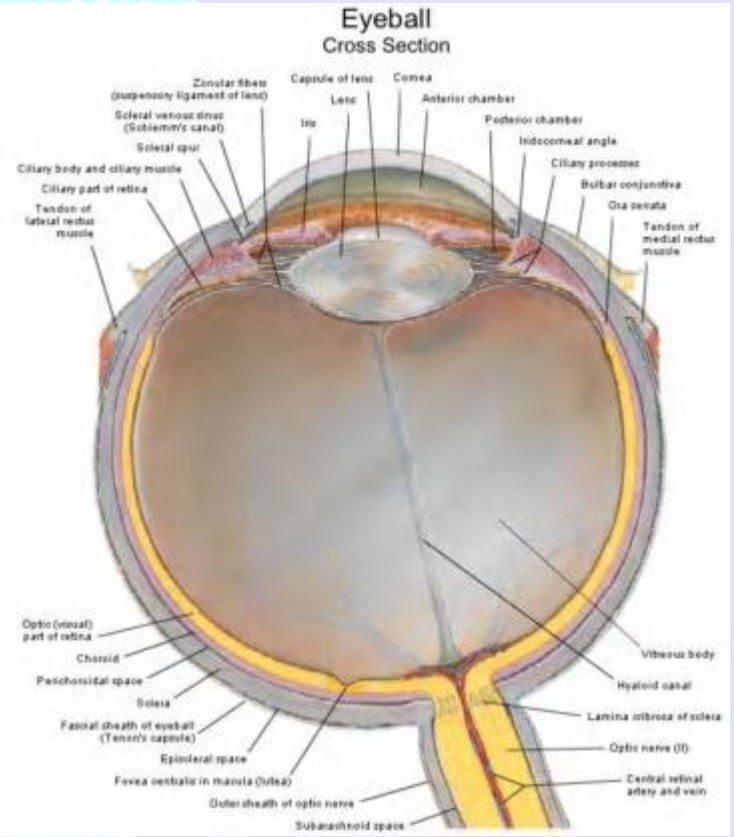
SCLERA

- **Posterior 5/6** of the external or fibrous coat
- Composed of **dense fibrous tissue**
- **Opaque** and **white** in colour
- Thickness about **0.5** mm
- Help to maintain the **size** and **shape** of the eye ball



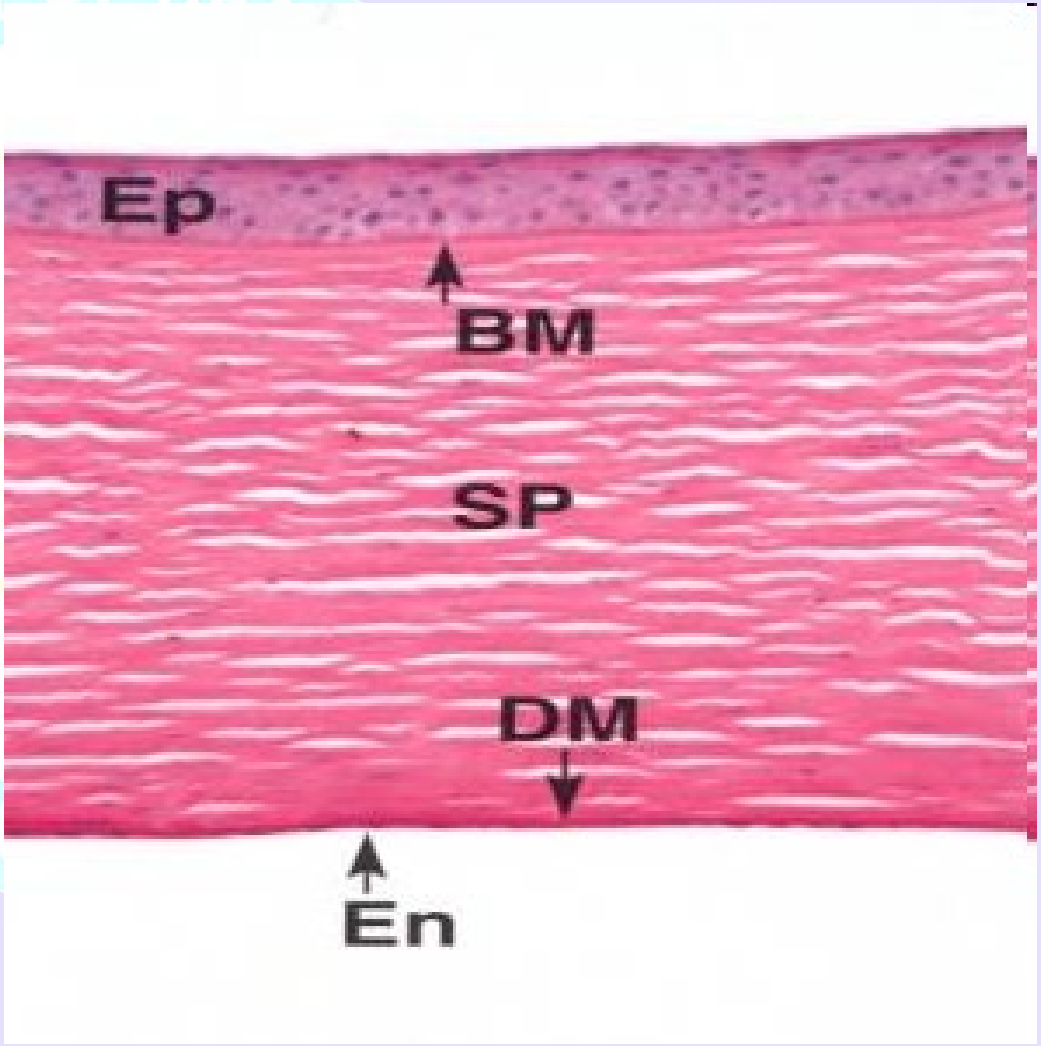
CORNEA

- **Anterior 1/6th** of the fibrous coat.
- Colourless and transparent.
- Thickness about **0.8 mm** in the center and **1.0 mm** at the periphery.
- Cornea is an **avascular** structure



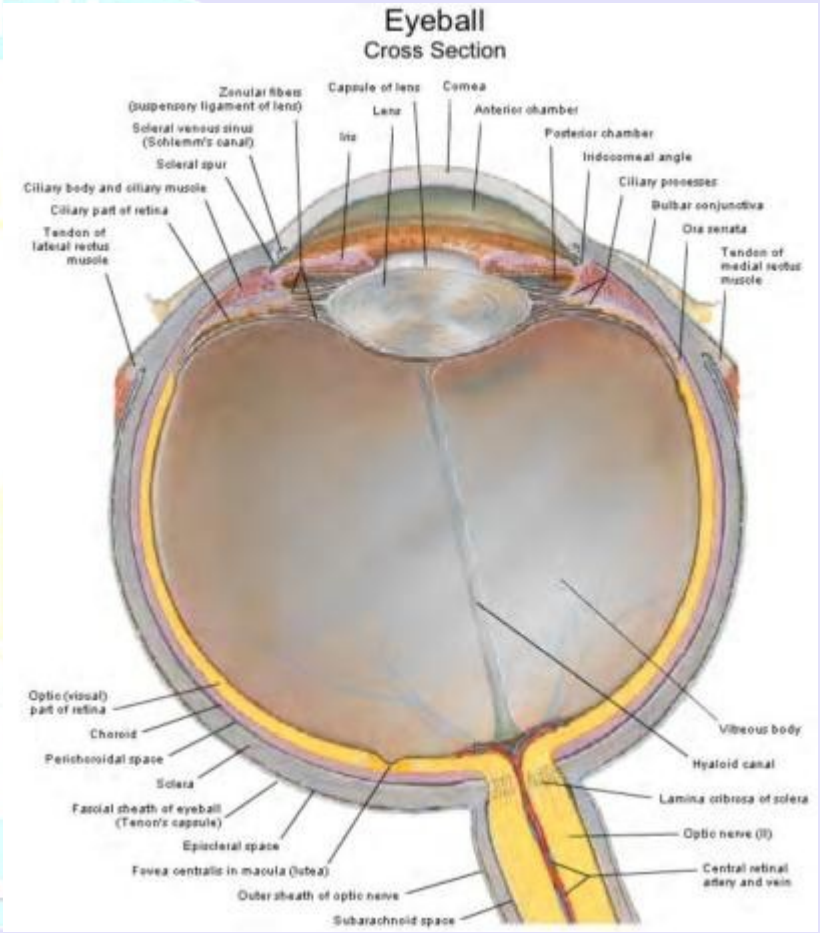
CORNEA T/S

- In transverse section (from before backwards) it composed of following five layers:
- **1.** Epithelium
- **2.** Bowman’s membrane
- **3.** Substantia propria
- **4.** Descemet's membrane
- **5.** Endothelium



CHOROID

- Forms the posterior part of uvea or uveal tract
- Externally it is separated from the sclera by a potential space called “perichoroidal space.”
- Structurally it composed of 4 layers: (from before backwards)
- **1.** Suprachoroid
- **2.** Vessel layer
- **3.** Capillary layer
- **4.** Lamina vitrea

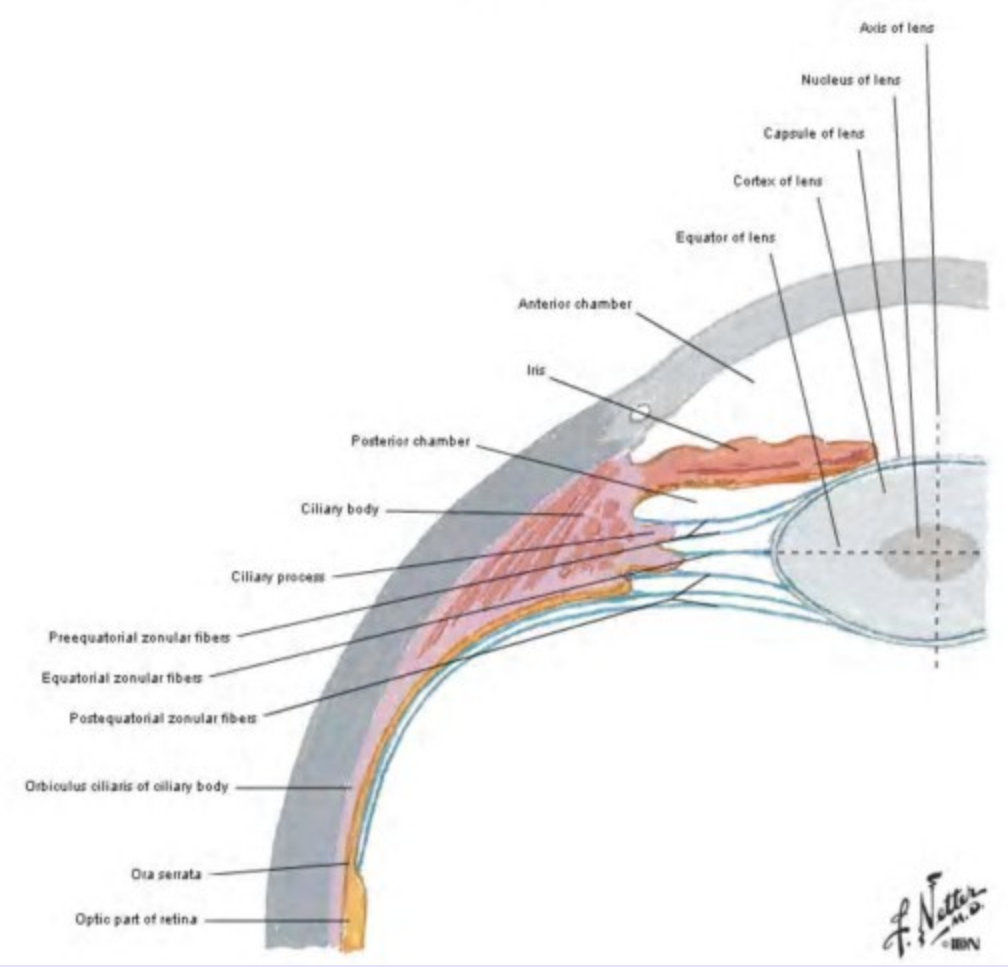


CILIARY BODY

- Choroid extends anteriorly as far as the ora serrata (anterior margin of the sensory portion of retina) anterior to the ora serrata, uveal tract thickened to form ciliary body
- In L/S appears triangular in shape:
- Base faces the anterior chamber
- Outer surface blends with sclera
- Inner surface faces vitreous body

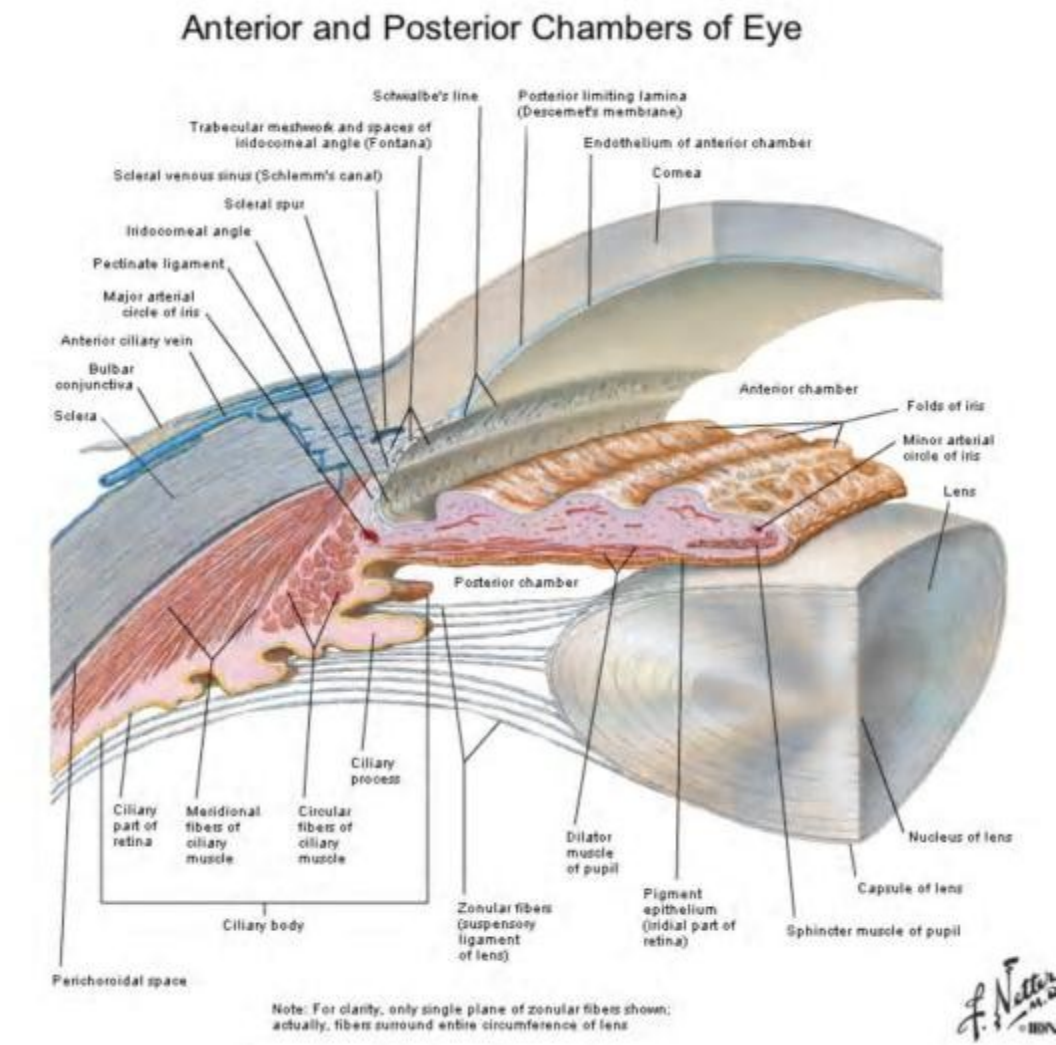
CILIARY BODY

- On naked eye inner surface show 2 zones:
- Posterior 2/3rd is smooth and darkedly pigmented called “pars plana”.
- Anterior 1/3rd is pale and bears about 70 – 80 radially arranged ridges (ciliary processes) called “pars plicata.



AQUEOUS HUMOR

- It is a watery fluid, has an inorganic ion composition similar to that of plasma, but contains less than 0.1% protein (plasma has about 7% protein)
- It has a high concentration of ascorbic acid, free amino acids, and sodium, chloride,& bicarbonate ions
- Glucose & urea levels are also lower than those of plasma

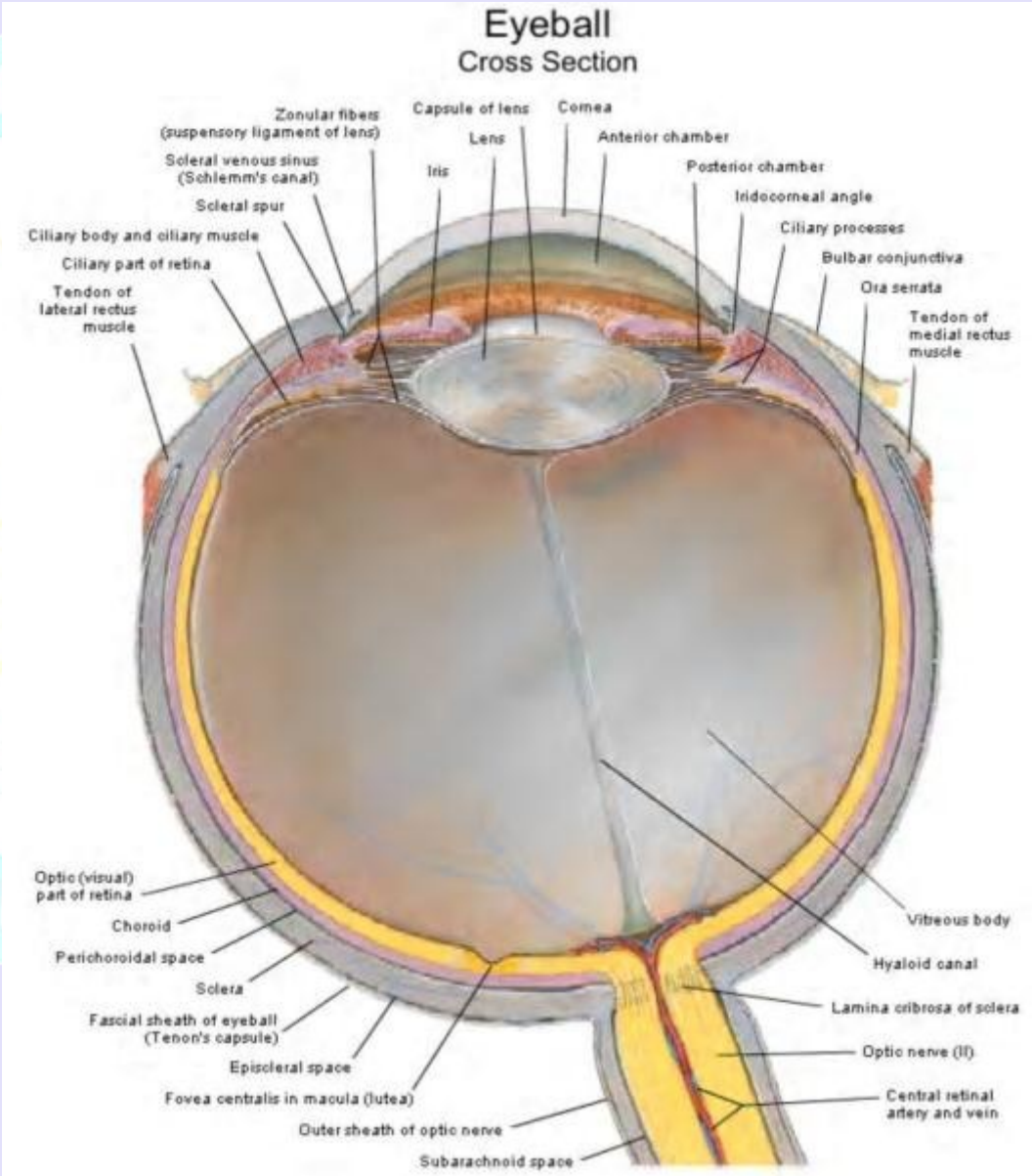


IRIS

- A thin circular, pigmented diaphragm
- Suspended in aqueous humor b/w cornea and lens.
- It has a round distensible aperture in the center called “Pupil”.
- Ciliary margin.
- Pupillary margin.
- Anterior surface devoid of epithelium - velvety appearance.

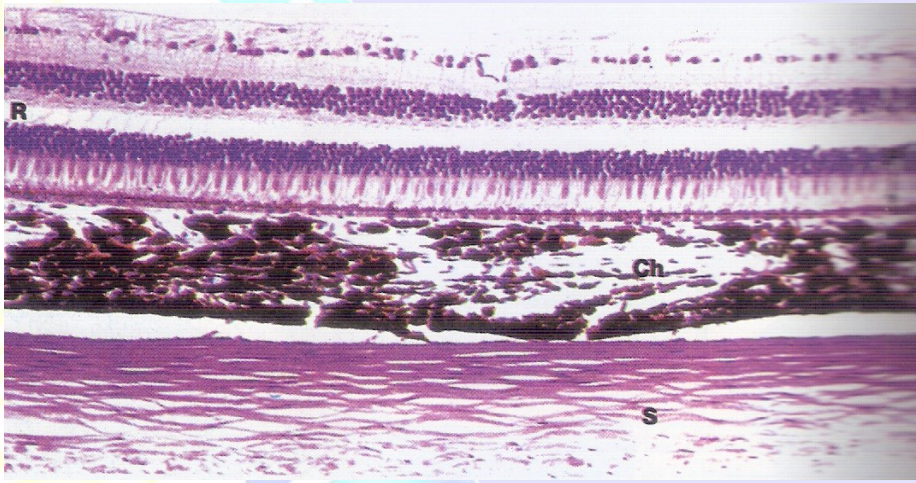
VITREOUS BODY

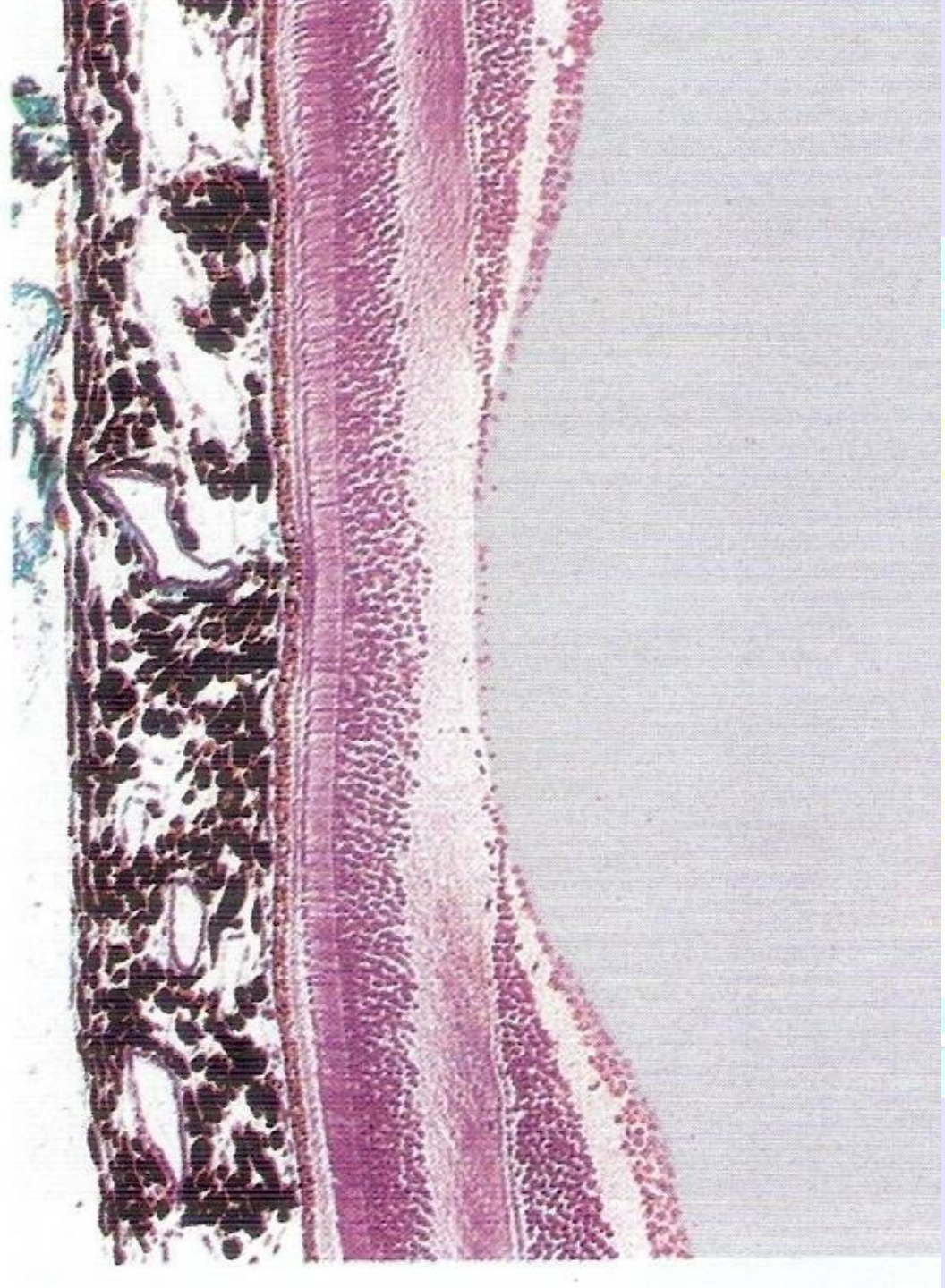
- Occupies the region of the eye behind the lens.
- A transparent gel that consists of water (about 99%),a small amount of collagen heavily hydrated with hyaluronic acid molecules and very few cells (hyalocytes).
- Hyalocytes synthesize collagen and hyaluronic acid.



RETINA

- It is the innermost layer of the eyeball
- It transduces the stimulus of light into nerve impulses, resulting in the sensation of vision
- On the bases of structural features the retina is divided into **ten layers**





FOVEA CENTRALIS

- A shallow, circular depression lying at the posterior pole of the optical axis of eye.
- Caused by the absence of inner layers of retina.
- Bipolar and ganglion cells accumulate in the periphery of depression, so that its center consists only of cone cells.
- In this area, blood vessels do not cross over the photosensitive cells.
- Light falls directly on cones in the central part of fovea, which helps account for extremely precise visual acuity of this region.

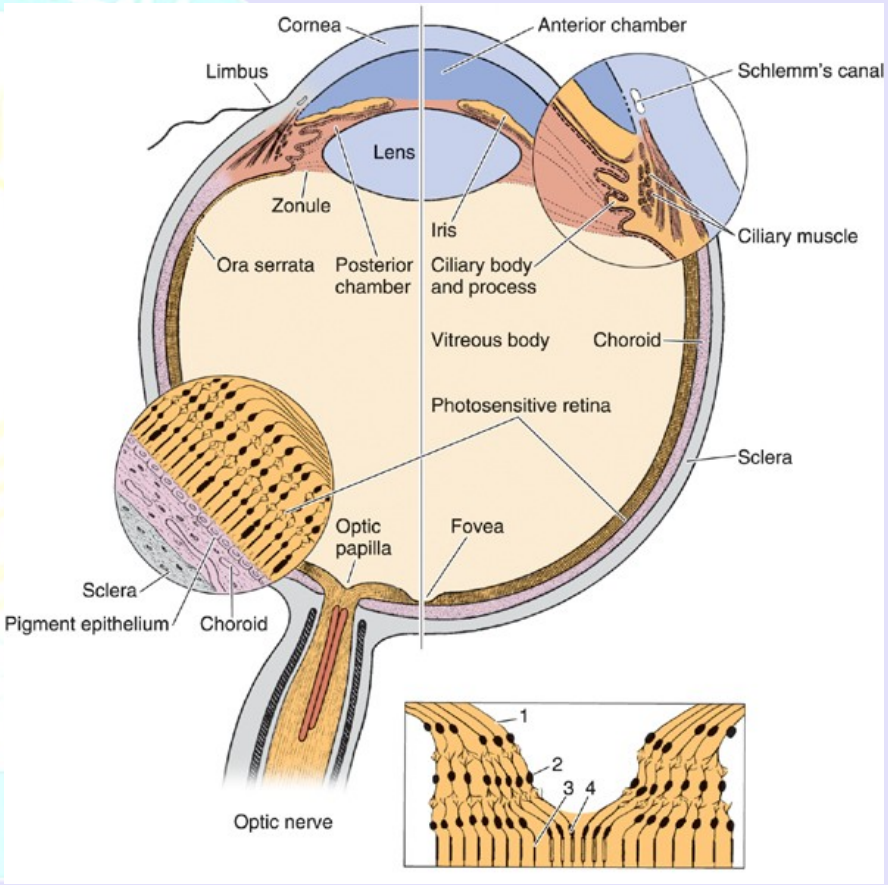
EYE LENS

- It is a transparent, biconvex body situated between the iris and vitreous body.

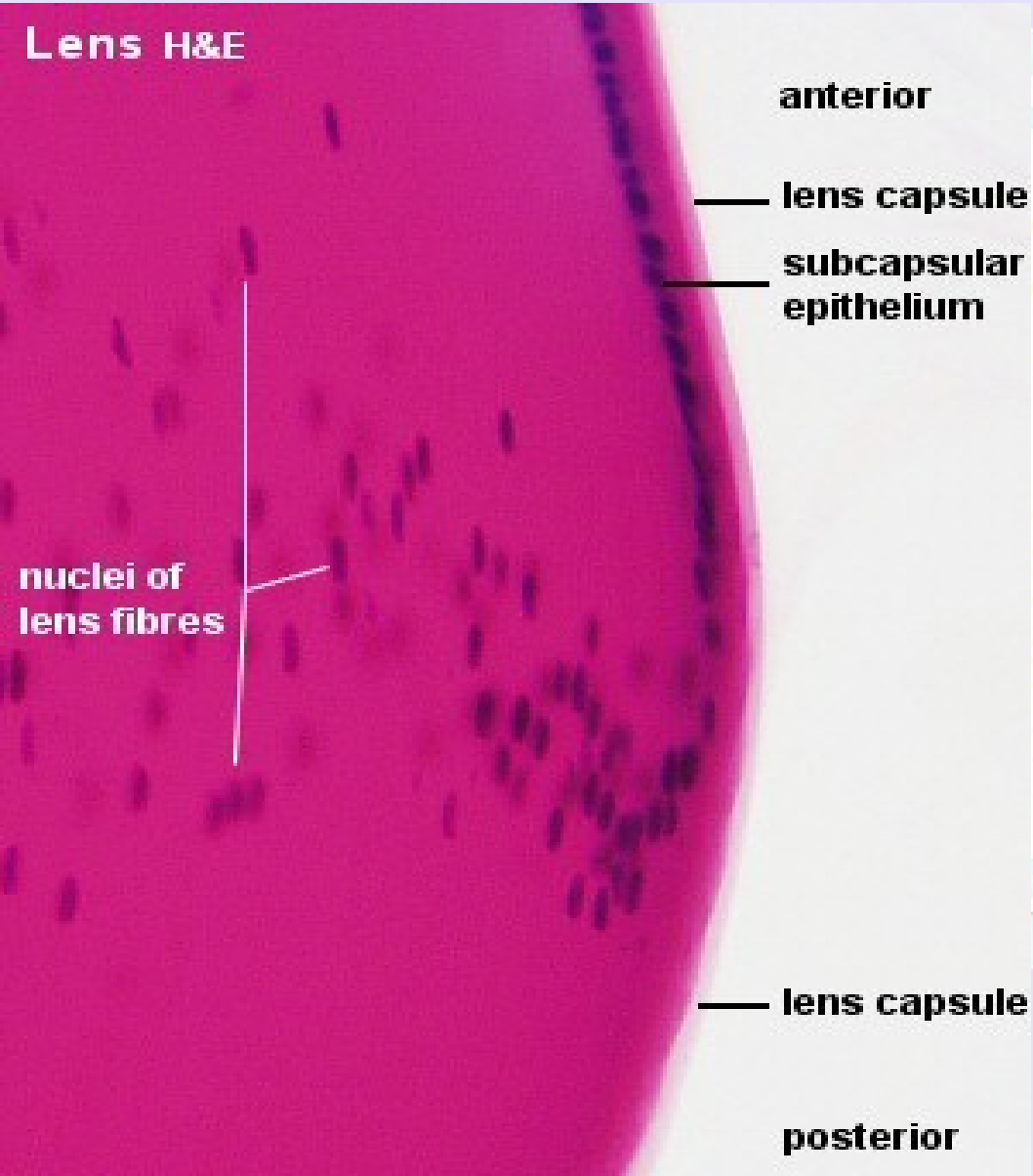
- It is elastic in the young subjects, and becoming harder with the age.

- **Structure:** It has three principal components

- Lens capsule.
- Subscapular epithelium.
- Lens substance.



EYE LENS



1. Lens capsule

- It envelops the lens
- It is 10 – 20 µm thick, homogeneous retractile membrane
- It is elastic
- It is composed of collagen type IV and glycoprotein

EYE LENS

2. SUBCAPULAR EPITHELIUM

- It is a simple cuboidal epithelium, present only on the anterior surface of the lens just under the capsule.
- Towards the equator, epithelial cells increase in height and transform into lens fibers.
- Lens grows throughout life by the addition of these fibers

EYE LENS

3. Lens Substance

- It consists of elongated prismatic lens fibers.
- Lens fibers are highly differentiated cells, derived from cells of the subscapular epithelium.
- Lens fibers eventually lose their nuclei and other organelles and become greatly elongated, attaining dimensions of 7 – 10 mm in length, 8 – 10 µm in width, and 2 µm in thickness.



THANKYOU